

What robot pretraining buys a vision-language-action model

A controlled component-interaction study across two embodiments

Six arms • one frozen-backbone head • 2,600 closed-loop rollouts • LIBERO-Goal and ALOHA transfer-cube

Existing work studies how to choose data and how to choose a backbone. This study asks what happens *between* the components: how backbone pretraining, camera configuration, action space and task structure interact to decide whether a policy actually succeeds in the loop.

The six claims

#	Claim	Key number
C1	A camera matters far more than a pretrained backbone	+29.0 vs +2.5 pts
C2	Pretraining helps on one task and not the other, at every stage of training	+9.5 pts, $p = 0.0067$
C3	The whole advantage sits in one transition	$p = 0.0009$
C4	The stock policy never opens the receiving gripper — and offline metrics miss it	26.2% vs 14.5%
C5	Language ablation has a floor that must be measured	1.14–1.24×
C6	Visual reliance tracks what the action is defined relative to	0.038 vs 0.11

Setup

One identical 19.2M-parameter head — token adapter plus DiT flow-matching decoder — is trained against **frozen** published backbones and compared by closed-loop rollout. Every arm shares the head, the read depth, the optimiser schedule and the evaluation protocol, so a difference in success rate is attributable to the factor that moved.

The pair spans one lineage: stock **Qwen3-VL-2B** → Cosmos-Reason2-2B → **GR00T N1.7**. Both hops are verified real finetunes (584/625 and 476/493 tensors differ), so the comparison covers the whole robot-pretraining treatment rather than one weak hop.

Two controls make the pair comparable. **Layer matching:** both arms are read at layer 16 — GR00T's own `select_layer`, intermediate for Qwen3-VL's 28 layers — because reading each at its own last layer would compare depth 16 against depth 28 and attribute a depth effect to pretraining. **Final-norm correction:** since layer 16 is final for one arm and intermediate for the other, the language stack's final RMSNorm is applied to the intermediate read, without which the pair would differ by normalisation rather than by weights.

C1 · A camera matters far more than a pretrained backbone

CLAIM 1

On single-arm manipulation, observation configuration outweighs backbone pretraining by an order of magnitude.

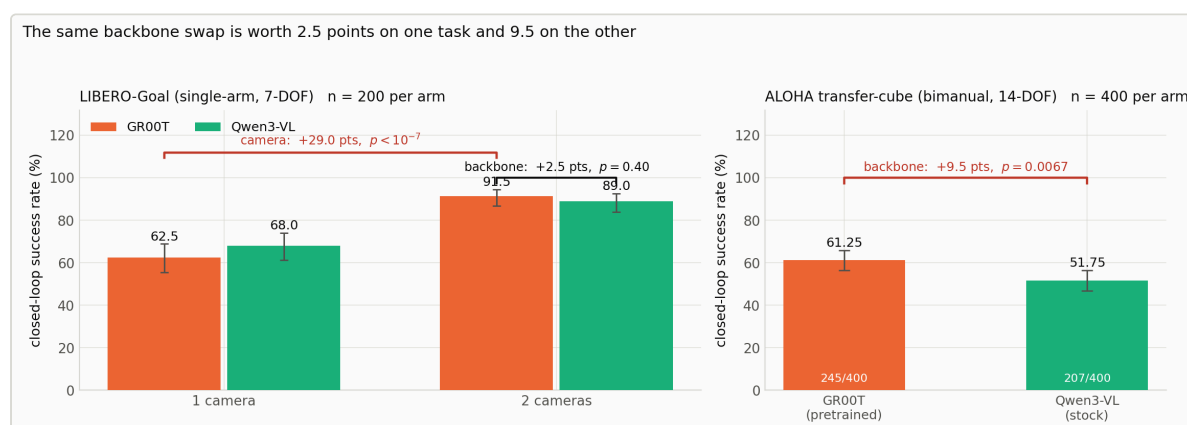
EVIDENCE

Arm	Backbone	Cameras	Success (n = 200)
exp03	GR00T N1.7	1	62.5%
exp04	Qwen3-VL-2B	1	68.0%
exp05	GR00T N1.7	2	91.5%
exp06	Qwen3-VL-2B	2	89.0%

Adding the wrist camera: **+29.0 points** (GR00T) and **+21.0 points** (Qwen3-VL), $p < 10^{-7}$. Swapping the backbone at the benchmark's own two-camera specification: **+2.5 points**, $p = 0.40$.

Supporting detail: the instruction is load-bearing for every arm (0/200 under a swapped instruction, all ten tasks), and both two-view arms converge to the same routing despite different backbone weights — wrist attention 51.6% vs 54.5%, ablation cost $\times 11.94$ vs $\times 11.64$.

FIGURE



The study in one figure. Left: on LIBERO-Goal a second camera is worth +29.0 points while swapping the backbone at that same observation spec is worth 2.5 ($p = 0.40$). Right: on bimanual ALOHA the identical backbone swap is worth +9.5 points ($p = 0.0067$). Error bars are Wilson 95% intervals.

WHY THE BACKBONE COMPARISON IS A NULL HERE

A 2.5-point aggregate gap is compatible with two different stories: the arms converged to behaviourally equivalent policies, or they behave differently and the differences cancel. Both testbeds are paired episode-for-episode — LIBERO by its 50 fixed initial states, ALOHA by seed — so the two can be told apart.

	Both succeed	Both fail	Only pretrained	Only stock
LIBERO 2-view	165	4	18	13
ALOHA (for contrast)	73	55	47	25

Three observations, in decreasing order of weight. **(1) Little headroom remains** — both arms sit at 89.0–91.5% while oracle-action replay through the same harness returns 90–92%. That is not a hard ceiling (the two-view arms beat the replay rate on two tasks) but the task is largely solved from the two camera views. **(2) The head reaches the same solution from either backbone** — wrist attention 51.6% vs 54.5%, wrist-ablation cost $\times 11.94$ vs $\times 11.64$, text ablation $6.12\times$ vs $5.39\times$, PE sensitivity 0.111 vs 0.097. **(3) What remains is idiosyncratic, not systematic** — only 4 of 200 episodes defeat both arms, and 31 of the 35 failures are disjoint. ALOHA is the opposite: 55 shared failures, a real pool of hard scenes.

What the null does not mean. The aggregate 2.5 points conceals per-task differences that cancel — task 9 favours the pretrained arm by 25 points, task 0 favours the stock arm by 20, four tasks are exactly tied, and the per-task correlation is only $r = 0.471$.

These per-task gaps are *not resolved*: at $n = 20$ per task the standard error on a difference is 9.4 points, so ± 20 is about 2 SE and several excursions that size are expected across ten tasks by chance. The defensible statement is that the arms' differences **do not accumulate in one direction** — not that they behave identically task by task.

Bounded to: one task suite (LIBERO-Goal) and one backbone pair. The single-view arms are reported for completeness but carry no claim — single-view puts the pretrained arm outside its trained input configuration while leaving the stock arm inside its own, an asymmetry pointing the same way as the result.

The headroom argument means this null is *partly a property of the benchmark*, not only of the backbone: a harder single-arm task with more room above 90% could still reveal a backbone effect. Cohen's kappa is computed by the analysis script (LIBERO +0.121, ALOHA +0.283) but carries no claim here — with both arms near 90%, "both fail" is four episodes and kappa swings on a handful of cases. The raw 2×2 counts are the evidence.

C2 · Pretraining's payoff is task-specific, at every stage of training

CLAIM 2

Robot pretraining is not a general sample-efficiency prior: it is decisive on one task and absent on the other, from the first checkpoint to the last.

EVIDENCE — THE BIMANUAL TASK

ALOHA transfer-cube was chosen as the pre-registered falsifier for C1's null. Bimanual manipulation is inside GR00T's pretraining distribution, so the venue is biased *toward* finding an effect; a null here would have been strong evidence. It did not return a null.

Arm	Run 1	Run 2	Pooled (n = 400)	Wilson 95% CI
GR00T N1.7	60.0%	62.5%	61.25%	[56.4, 65.9]
Qwen3-VL-2B	49.0%	54.5%	51.75%	[46.9, 56.6]

+9.5 points, $z = 2.71$, $p = 0.0067$ — clears the Bonferroni bar for the six comparisons in this study ($0.05/6 = 0.0083$), and the intervals do not overlap. The two runs used disjoint seed ranges, so this is a genuine replication rather than a resampling of the policy's own noise.

EVIDENCE — THE WHOLE TRAINING CURVE

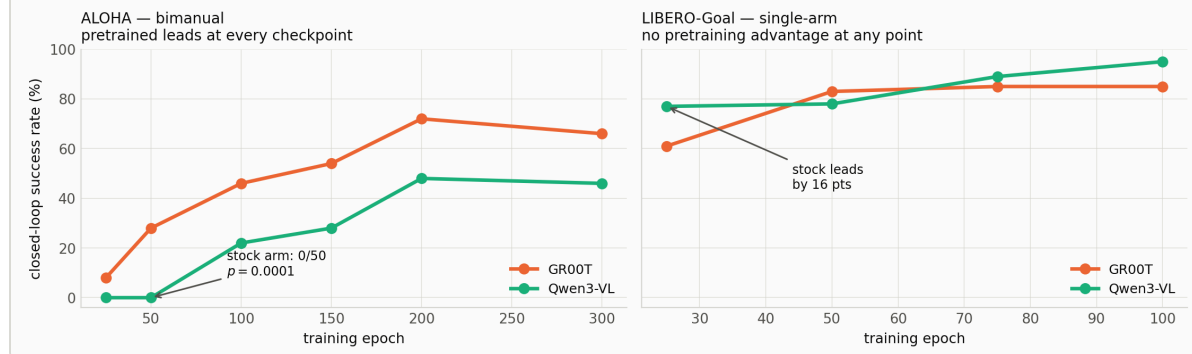
ALOHA epoch	25	50	100	150	200	300
GR00T	8.0%	28.0%	46.0%	54.0%	72.0%	66.0%
Qwen3-VL	0.0%	0.0%	22.0%	28.0%	48.0%	46.0%

LIBERO epoch	25	50	75	100
GR00T		61.0%	83.0%	85.0%
Qwen3-VL		77.0%	78.0%	89.0%

On ALOHA the stock backbone **cannot do the task at all** before epoch 100 — 0/50 at epochs 25 and 50 while the pretrained arm is at 28% (McNemar $p = 0.0001$) — and needs $\sim 2\times$ the epochs to reach any given rate. On LIBERO the *stock* arm leads at epoch 25 by 16 points. The early-training effect, the largest in the study, does not transfer.

FIGURE

Robot pretraining is not a general sample-efficiency prior



Both checkpoint ladders on a common axis. Left: on bimanual ALOHA the pretrained arm leads at every checkpoint. Right: on single-arm LIBERO-Goal it leads at none. Paired seeds throughout; LIBERO's 50 fixed initial states make its pairing exact.

Bounded to: the two testbeds differ on *six axes simultaneously* — instruction variation (10 vs 1), degrees of freedom (7 vs 14), arm count, action space, camera count, and distribution match to GR00T's pretraining corpus. The DOF hypothesis, the bimanual hypothesis and the distribution-match hypothesis all predict what was observed, and this design cannot separate them. Naming bimanual coordination as *the* cause is an interpretation, not a result.

Two further cautions: on LIBERO neither nominally significant point ($p = 0.0195, 0.0213$) survives Bonferroni over four comparisons, so the claim there is the negative one. And both ladders disagree with their own headline levels — read the ladders for *shape*, the pooled runs for *magnitude*.

C3 · The whole advantage sits in one transition

CLAIM 3

Pretraining does not raise general competence — it raises one conditional probability.

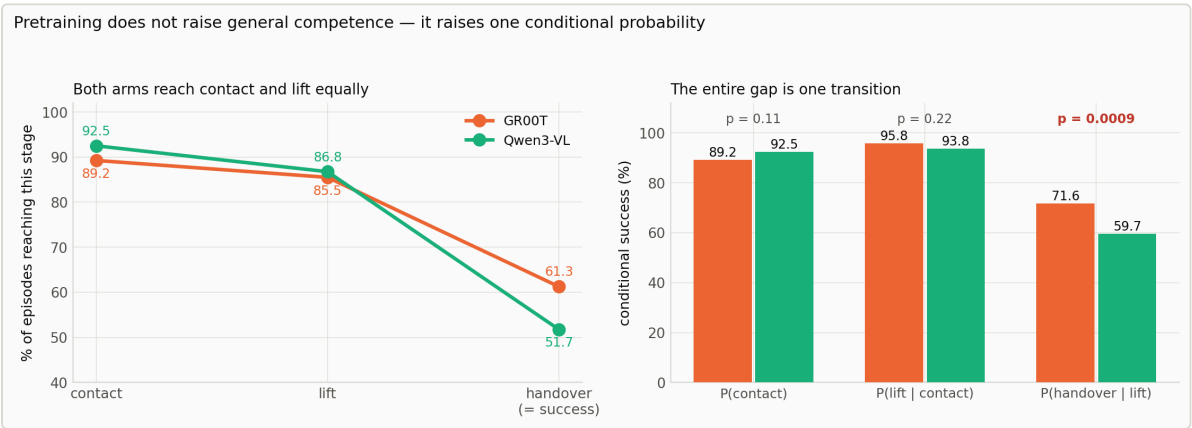
EVIDENCE

ALOHA scores contact / lift / handover / success. `max_reward == 3` never occurs in 800 episodes, so once the receiving gripper touches the cube the episode always completes, and the ladder is contact → lift → handover.

Stage	GR00T	Qwen3-VL	Δ	p
P(contact)	89.2%	92.5%	-3.3	0.11
P(lift contact)	95.8%	93.8%	+2.0	0.22
P(handover lift)	71.6%	59.7%	+12.0	0.0009

Both early stages run slightly *against* the pretrained arm, and the localisation is tighter ($p = 0.0009$) than the top-line result ($p = 0.0067$). Decomposing success along the environment's own reward ladder converts an aggregate difference into a mechanism at no additional rollout cost — a method other closed-loop studies can adopt directly.

FIGURE



Left: both arms reach contact and lift at the same rate — the curves separate only at the final transition. Right: the same data as conditional probabilities. P(handover | lift) carries the entire gap.

Bounded to: one task. Whether the pattern generalises to other multi-stage manipulation tasks is untested.

C4 · The stock policy fails to initiate the receive

CLAIM 4

Offline metrics cannot rank these policies — and the instrumented rollouts show what is physically different: the stock policy never opens the receiving gripper.

EVIDENCE — THE BLINDNESS

Measure (stock – pretrained, % of pretrained)	Δ
velocity loss (overall)	+0.1%
velocity loss (handover phase)	+1.9%
open-loop action error (nMAE, 14 dims)	–0.4%
PE sensitivity	+1.1%
attention mass on image	–5.2%
closed-loop success rate	–15.5%

Four accuracy measures differ by under 2% against a gap the rollouts resolve at $p = 0.0067$. The training objective *does* register that the handover window is hard — mid-phase loss is 1.8× the late-phase value for both arms — but not that one arm is worse at it.

EVIDENCE — THE REASON

Instrumented rollouts ($n = 200$ per arm) log per-step 14-DOF state, commanded action, the reward timeline and per-replan attention. Arm identity was *measured*, not assumed: the picking arm is whichever block moves first in successful episodes. Classifying episodes that lifted the cube but failed the handover:

The receiving gripper rests **closed** (~ 0.19) and opens to ~ 0.96 — verified from the trajectories, not assumed. The receive sequence is **open** → **cube arrives** → **close on it**, so the three ways it can fail are: never open, open but never close, or close on nothing. As a rate over all episodes that reached the lift:

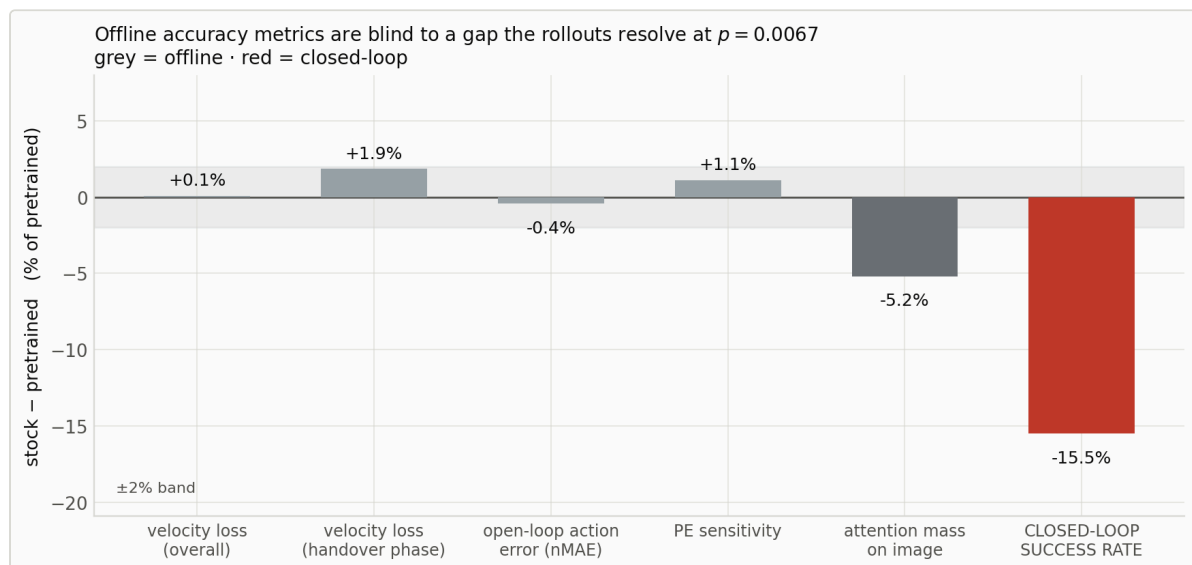
Failure mode	GR00T	Qwen3-VL	p
receiving gripper never opens	14.5%	26.2%	0.0068
opened but never closed	13.3%	9.3%	0.24
closed on nothing	2.3%	4.7%	0.24

One failure mode carries the whole gap. The stock arm's excess of never-opened episodes is **+11.7 points**, against a total P(handover | lift) gap of **11.9 points**. The other two modes run slightly the other way.

The stock policy does not aim badly, mistime the grasp, or drop the cube more often. It **fails to initiate the receive at all** — the receiving gripper stays closed while the episode times out. That is why per-step action error is blind to it: one dimension of fourteen, held at a plausible constant, in a chunk whose other thirteen dimensions are fine.

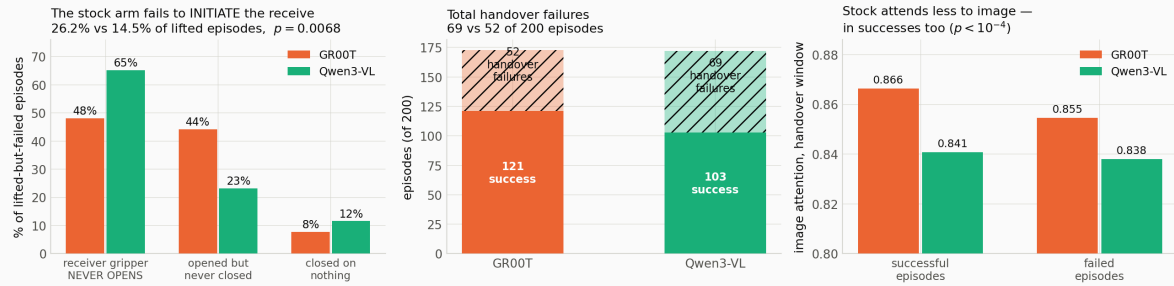
Attention was recorded **during rollout**, not on dataset frames: dataset frames are successful expert demonstrations and cannot be linked to the policy's own failures. Over the five replans after the lift the stock arm allocates less mass to image tokens — 0.8407 vs 0.8663 in successful episodes (Welch $t = 9.01$) and 0.8380 vs 0.8546 in failed ones ($t = 5.24$), both $p < 10^{-4}$. The cue to open the receiving gripper is *visual*, and the stock arm weights that channel less in exactly that window — a coherent reading of two measurements, not a demonstrated causal chain.

FIGURES



Every offline diagnostic as a between-arm difference, on the same axis as the closed-loop result. Four accuracy measures sit inside the $\pm 2\%$ band.

The stock policy fails to initiate the receive — it never opens the receiving gripper



Instrumented rollouts. Left: the stock arm's failures concentrate in one mode -- the receiving gripper never opens. Centre: total handover failures. Right: image attention during the handover window, split by outcome so the comparison cannot be an artifact of failed episodes running longer.

Bounded to: these are correlational measurements on frozen checkpoints; they locate where two policies differ but cannot prove the difference causes the gap. Because the attention difference appears in successful episodes too, it is a stable policy-level habit, *not* a signature of impending failure; it must not be described as the policy recognising anything.

Correction on record: an earlier version of this analysis reported "premature receiver close" as the dominant failure for both arms (59.6% vs 59.4%, $p = 0.98$) and concluded the failure mix was identical. That was an artifact — the classifier had the gripper polarity backwards and tested for "closed most of the post-lift window", which is the gripper's default state (closed 84% of all timesteps), so it fired for nearly every episode of both arms. The corrected classification above finds the mixes differ.

CLAIM 5

Zeroing text tokens costs something even when the text carries no information — so ablation ratios must be read against that floor, not against 1.0.

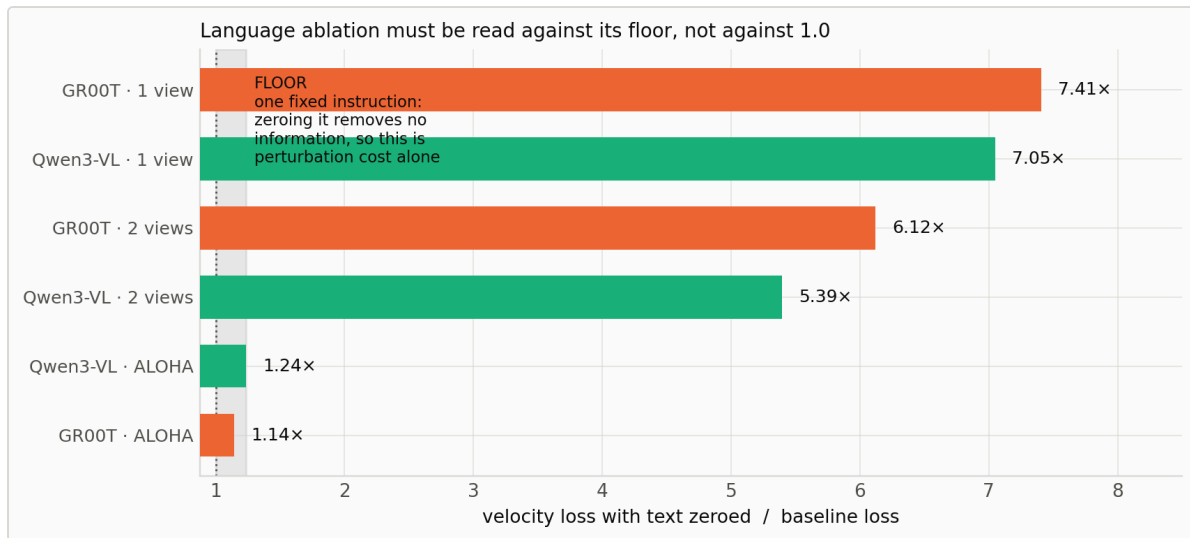
EVIDENCE

ALOHA has a single fixed instruction, so zeroing its text tokens removes no task-discriminative information. Loss nonetheless rises **1.14×** (GR00T) and **1.24×** (Qwen3-VL). That residual is the cost of an off-distribution perturbation alone — a control condition an ablation on a multi-instruction dataset cannot supply for itself.

Arm	Instructions	Ablation ratio
GR00T · ALOHA	1 (fixed)	1.14× — <i>floor</i>
Qwen3-VL · ALOHA	1 (fixed)	1.24× — <i>floor</i>
Qwen3-VL · 2 views	10	5.39×
GR00T · 2 views	10	6.12×
Qwen3-VL · 1 view	10	7.05×
GR00T · 1 view	10	7.41×

A second reading falls out of the same table: the two-view arms depend on language measurably *less* than the one-view arms (5.39×/6.12× against 7.05×/7.41×), consistent with a wrist camera recovering information the instruction previously had to supply.

FIGURE



Text-ablation ratio across all six arms. The grey band is the floor measured on ALOHA's single fixed instruction — the cost of the perturbation alone.

Bounded to: the floor is measured on one dataset and one head; its magnitude on other setups is unknown. The methodological point — that a floor exists and is not 1.0 — does not depend on its exact value.

CLAIM 6

A policy leans on image position when the action is a displacement it cannot infer from one frame, and not when the action is a destination the frame already implies.

EVIDENCE

Zeroing the 2D positional encoding shifts actions by **0.097–0.111** under end-effector control but only **0.038** under 14-DOF joint-space control with full proprioception.

The obvious objection is that ALOHA's scene is simply visually simpler. A gradient or Jacobian analysis cannot settle that — an impoverished scene produces low image sensitivity under either explanation. A probe of what the tokens *contain* can:

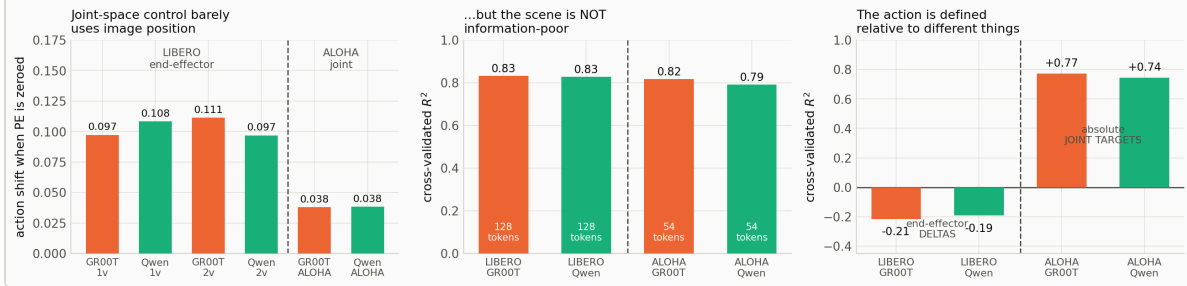
Arm	Image tokens	R ² (arm state)	R ² (action)
LIBERO GR00T 2-view	128	0.832	−0.215
LIBERO Qwen3-VL 2-view	128	0.828	−0.192
ALOHA GR00T	54	0.817	0.771
ALOHA Qwen3-VL	54	0.791	0.743

Arm configuration is **equally readable on both testbeds** (0.830 vs 0.804) — and ALOHA reaches that from **54 image tokens against LIBERO's 128**. The scene is not information-poor, so "visually simpler" does not explain the lower PE sensitivity.

The second column identifies the operative variable. LIBERO commands end-effector *deltas*, which one frame does not determine (R² −0.20); ALOHA commands *absolute joint targets*, which sit close to the pose the image already encodes (R² +0.76). The driver is what the action is defined relative to — not degrees of freedom, not arm count.

FIGURE

Visual reliance tracks what the action is defined relative to, not scene complexity



Left: PE sensitivity across all six arms. Centre: arm configuration is equally readable from image tokens on both testbeds, and ALOHA does it with fewer tokens. Right: action recoverability from a single frame, the variable that separates them.

Bounded to: the action-recoverability contrast is *partly definitional* — absolute targets are near the current state by construction — so it is reported as *explaining* the PE result rather than as independent evidence for it. Token geometry also differs between the testbeds (54 vs 128) and is not separately controlled.

Validity

The harness is correct. With the policy removed from the loop and the demonstrations' own actions replayed through the same env construction, success detector and step budget, LIBERO replays at 90.0% (no settling) and 92.0% (5 settling steps) — the band published methods report, so this is not a stricter harness. It is a replay rate, not a policy ceiling: the two-view arms exceed it on two tasks.

Training budget is not the constraint. Each arm takes 53,760 optimizer steps at batch 128 — 7.2× more samples and ~34× more passes over the data than a published LeRobot-family recipe on LIBERO. Validation flattens after epoch ~75 and open-loop action correlation reaches 0.979–0.997.

Instrumentation did not perturb the policies. The traced run reproduced 60.5% / 51.5% against the original 60.0% / 49.0% on the same seeds.

What this study cannot settle

- **Two backbones, one lineage.** Every backbone claim rests on one pair.
- **The six-axis confound** (see C2) — two testbeds cannot separate six covarying differences.
- **One head architecture.** "Backbone barely matters on LIBERO" may be specific to a head with this capacity and cross-attention design.
- **Correlational diagnostics.** A causal test would swap handover-window tokens between arms.
- **Simulation only.** No physical robot.
- **Nulls are not equivalence.** At $n = 200$ per LIBERO condition the standard error is 3.3 points.

Summary

Robot pretraining of a frozen VLM backbone is **not a general sample-efficiency prior**. Its payoff is task-specific, and where it fails it fails at every point in training rather than only at convergence.

On single-arm pick-and-place at the benchmark's own two-camera specification it contributes nothing measurable — 2.5 points at $p = 0.40$, and no advantage at any checkpoint — while a second camera contributes +21 to +29. On bimanual handover it is the difference between 0% and 28% success at 50 epochs, halves the epochs to any target, and leaves a replicated +9.5-point advantage that localises to a single transition.

Throughout, the offline metrics these systems are trained on and selected by failed to rank the resulting policies — and the instrumented rollouts show what they were missing: the stock policy never opens the receiving gripper, one dimension of fourteen held at a plausible constant while the other thirteen behave normally.

Full design, statistics and defect log: `RESULTS.md` • Framing and component ledger: `OBJECTIVE.md` • Figures: `scripts/analysis/plots_paper.py`